

Kelan S. Zielinski

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Work & Leadership Experience

Motorola Solutions Inc.

Plantation, FL

Electrical Engineering Intern

May 2024 - August 2024

- Collaborated with RF engineers to analyze and adjust circuit parameters, successfully achieving optimal network impedance using a Smith chart.
- Implemented automated tests in LabVIEW to verify network matching, comparing RSSI values to predetermined benchmarks, ensuring signal integrity.
- Validated scattering (S2P) parameters on a Vector Network Analyzer (VNA) and redesigned the network on Advanced Design System (ADS) software.
- Performed root cause analysis using an oscilloscope to identify timing issues that resulted in unexpected power cycles between a radio and an uninterruptible power supply (UPS).

IoT-Driven Crop Production System

Coral Gables FL

Hardware Engineering Intern - University of Miami

January 2023 - May 2024

- Designed and built an IoT-controlled hydroponics system to cultivate 2000+ plants per year within a footprint of less than 50 square feet.
- Developed a hardware prototype and validated components, including I2C and UART sensors and actuators, based on system requirements.
- Utilized an EDA simulation platform to model and optimize system circuitry for improved performance and reliability.

University Of Miami – Design For Testability Laboratory

Coral Gables, FL

Teaching Assistant

January 2024 - May 2024

- Aided students in designing and implementing a digital signal processor in VHDL.
- Provided assistance with debugging timing issues in a large system.

HKN Honor Society (IEEE) – University of Miami Chapter

Coral Gables, FL

Eta Kappa Nu – Honor Society Member

February 2024 - Present

- Supported students in mastering fundamental engineering concepts.
- Provided programming suggestions for HDL and C++ based projects.

Projects

FPGA-Based Quantum Emulator – VHDL, Python, ModelSim

May 2025

- Designed and implemented a quantum system emulator on an FPGA using VHDL, modeling the time evolution of qubits based on their Hamiltonian.
- Optimized FPGA resource utilization for efficient matrix exponentiation and state vector updates to mimic quantum dynamics.

Advanced Digital Signal Processor – VHDL, ModelSim

April 2024

- Implemented the ADSP-2100 digital signal processor to perform specific computations based on the instruction.
- Integrated Built-In-Self-Test (BIST) architecture to generate signatures and expose faults within the system design.

MIPS RISC CPU – VHDL, ModelSim, Quartus Prime

October 2024

- Programmed, simulated, and implemented a 32-bit RISC processor.

Rotor-based Encryption Machine – Verilog, ModelSim, Quartus Prime

November 2023

- Developed a Verilog-based model of the Enigma M3, featuring three dynamic rotors with rotational functionality and a reflector component.
- Verified data transfer and timing aspects of system design using RTL viewer and waveforms.

Technical Skills

Technical Skills: VHDL, Verilog, C++, C, Python, ARM Assembly, Soldering, PCB Design & Testing

Developer Tools: ModelSim, ADS, Cadence, PSpice, MATLAB, Intel Quartus Prime, KiCad, Unix, LabVIEW, AutoCAD

Test Equipment: Oscilloscopes, Spectrum Analyzers, Logic Analyzers, Network Analyzers, Multimeters, Signal Generators

Education

University of Miami

Coral Gables, FL

Bachelor of Science in Electrical Engineering – GPA: 3.72

May 2025

- **Relevant Coursework:** Digital System Design & Testing, Discrete Time Signals & Systems, Computer Organization & Design, Circuits Signals & Systems, Design for Testability(DFT) Laboratory, Electronics, Logic Design, Structured Digital Design, Electromagnetic Field Theory, Data Structures